On Covering Spaces

A mini talk given on discord by .hchan

Sept 2026

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Primer in Point Set Topology

Topologies

Definition 1

Let X be a set. A *topology* $\mathcal{T}$ on X is a collection of subsets of X, called the *open sets*, such that the following three properties are satisfied:

1. $\varnothing, X \in \mathcal{T}$.
2. If $U_1, \ldots, U_n \in \mathcal{T}$ then so is $\bigcap_{i=1}^{n} U_i$; i.e. ($\mathcal{T}$ is closed under *finite intersection*).
3. If $\{U_\alpha\}_{\alpha \in \Lambda} \subseteq \mathcal{T}$ then so is $\bigcap_{\alpha \in \Lambda} U_\alpha$; i.e. ($\mathcal{T}$ is closed under *arbitrary union*).

For simplicity we will often refer to X itself as a topological space.

Example 2

If (X, d) is a metric space, then the set of open sets under d form a topology on X.

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Example 2

If (X, d) is a metric space, then the set of open sets under d form a topology on X.

Continuous functions

Definition 3

Suppose that $(X, \mathcal{T})$ and $(Y, \mathcal{S})$ are two topological spaces. A function $f : X \to Y$ is said to be *continuous* if the preimage of any open set in Y is an open set in X; i.e. if $U \in \mathcal{S}$ then $f^{-1}(U) \in \mathcal{T}$.

Theorem 4

When X and Y are metric spaces, this definition of continuous functions is equivalent to the $\epsilon - \delta$ definition you have seen in real analysis.

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Continuous functions (cont.)

Proof.

($\Rightarrow$) : Suppose $f : X \to Y$ is continuous under the preimage definition.
Fix $x_0 \in X$ and $\varepsilon > 0$. Then $B(f(x_0), \varepsilon) \subseteq Y$ is open, so $f^{-1}(B(f(x_0), \varepsilon))$ is as well by assumption. It also contains x_0, so there exists $\delta > 0$ with $B(x_0, \delta) \subseteq f^{-1}(V)$. But then this is exactly the ε-δ definition.

($\Leftarrow$) : Suppose $f : X \to Y$ is continuous under the ε-δ definition.
Because every open set is the union of open balls, it is sufficient to show that the preimage of a sufficiently small open ball is open.
Fix $x_0 \in X$ and $\varepsilon > 0$. By assumption there is $\delta > 0$ such that
$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon$. But then this is the same as saying
$B(x_0, \delta) \subseteq f^{-1}(B(f(x_0), \varepsilon))$. But then this implies that every point of $f^{-1}(B(f(x_0), \varepsilon))$ has an open neighbourhood contained in it, so it has to be open. □

Homeomorphisms

Definition 5

Suppose that X and Y are two topological spaces. A function $f : X \to Y$ is said to be a *homeomorphism* if it is bijective and bicontinuous.
In which case, X and Y are said to be *homeomorphic*.
This should be thought of as an isomorphism of topological spaces.

Example 6

Unlike the case in group theory, a bijective *continuous* function does not have to be a homeomorphism.
For a counter example, consider $f : [0, 2\pi) \to S^1$, $f(t) = (\cos(t), \sin(t))$. The set $[0, 1)$ is open in $[0, 2\pi)$ but $f([0, 1))$ is not; so even though f is continuous its inverse f^{-1} is not.

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Quotient spaces

Definition 7

Just like how we can take quotients of groups, it turns out that we can take quotients of topological spaces.
Suppose that $(X, \mathcal{T})$ is a topological space, and $\sim$ is some arbitrary equivalence relation on X. Then let $X/\sim$ denote the set of equivalence classes of $\sim$, and $\pi : X \to X/\sim$, be the natural projection.
The *quotient topology* $\mathcal{T}_\sim$ on $X/\sim$ is defined by

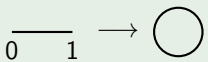
$$U \in \mathcal{T}_\sim \iff \pi^{-1}(U) \in \mathcal{T}.$$

$(X/\sim, \mathcal{T}_\sim)$ is called the *quotient space*, and π is the quotient map. It is the finest topology on $X/\sim$ for which π is continuous.

Geometric intuition[1]

Example 8

The circle. Identify the endpoints of an interval: $[0,1]/(0 \sim 1) \cong S^1$.



Example 9

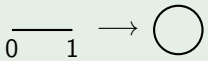
The torus. Identify opposite sides of a square, respecting arrow direction: $[0,1]^2/\sim \cong S^1 \times S^1$.

[1]

Geometric intuition[1]

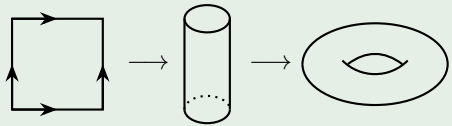
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[1]

Topological Covers

Topological covers

Definition 10

Let X be a topological space. A *cover* of X is a pair (Y, p), where Y is a topological space and $p : Y \to X$ is a continuous function; such that the following condition is satisfied:

> *For every $x \in X$, there exists an open neighbourhood U of x such that $p^{-1}(U)$ is homeomorphic to a disjoint sum of copies of U.*

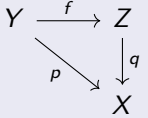
In general there may be any number of copies of U present, but in the context of this talk you may assume that there are at most countably infinitely many copies.

p is refered to as the projection map or the covering map.

Morphisms

Definition 11

Suppose that $p : Y \to X$ and $q : Z \to X$ are covers of X.
A *morphism* of covering spaces over X is a continuous function $f : Y \to Z$ such that the following diagram *commutes*:

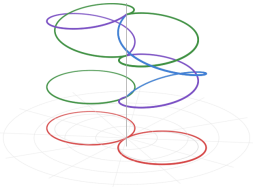


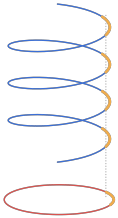
in other words, $p(y) = q(f(y))$ for all $y \in Y$. If f is further more a homeomorphism, then it is said to be an isomorphism.

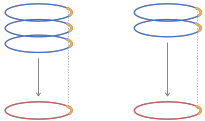
The collection of all covers of a space forms a *category*.

Some examples[2]









[2]Images generated by Sonnet 5.

Degree

Definition 12

Suppose that X is a *connected* space. Recall that this means that X cannot be written as the union of two disjoint nonempty open sets; or equivalently that the only nonempty clopen set of X is the entire space.

Then if $p : Y \to X$ is any cover, the *fiber* above every point has the same cardinality; i.e. the size of $p^{-1}(x)$ is the same for all $x \in X$. In which case, $|p^{-1}(x)|$ is defined to be the *degree* of the cover.

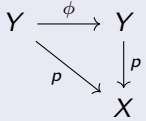
(Proof is left as Exercise 2).

Group Actions

Cover automorphisms

Definition 13

Suppose that $p : Y \to X$ is a cover. Then an *cover automorphism* of Y over X is an automorphism / isomorphism of Y in the category of covers of X (henceforth all automorphisms refer to cover automorphisms unless otherwise specified).
More concretely, it is a homeomorphism $\phi : Y \to Y$ such that the following diagram *commutes*:



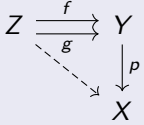
in other words, $p(y) = p(\phi(y))$ for all $y \in Y$.

The set of all cover automorphisms forms a group, denoted $\mathrm{Aut}(Y|X)$.

The critical lemma

Lemma 14

Let $p : Y \to X$ be a cover and Z, X connected topological spaces.
Suppose that f, g are two continuous maps such that $p \circ f = p \circ g$. If there exists $z \in Z$ with $f(z) = g(z)$, then $f = g$.



Corollary 15

An automorphism ϕ of a connected cover $p : Y \to X$ having a fixed point must be trivial (Take $Z = Y$ and $f = id$ and $g = \phi$ in the lemma above).

Proof.

Suppose that $y = f(z) = g(z)$. Take some connected open neighbourhood V of $p(y)$ satisfying the condition in the definition of a cover and let $U_i \cong V$ be the component of $p^{-1}(V)$ containing y.

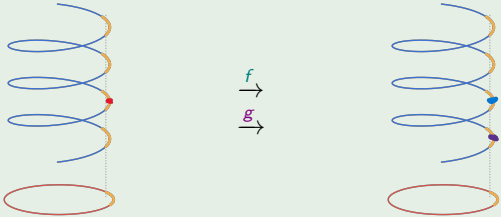
By continuity f and g must both map some open neighbourhood W of z into U_i. Since $p \circ f = p \circ g$ and p maps U_i homeomorphically onto V, f and g must agree on W.

The same argument can be reapplied to show that if $f(z') \neq z(y')$ for some point $z' \in Y$, f and g must map a whole open neighbourhood of z' to different components of $p^{-1}(V)$.

Thus the set $\{z \in Z : f(z) = g(z)\}$ is nonempty and clopen in Z, so by connectedness it is the whole of Z. □

A visual proof[3]

Example 16



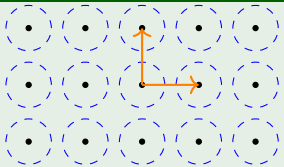
[3] Images generated by Sonnet 5.

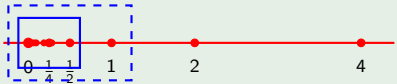
Even actions[4]

Definition 17

Suppose that Y is a topological space, and G is a group that acts on Y via continuous functions.
The action is said to be *even* if each point $y \in Y$ has some open neighborhood U such that the open sets gU are pairwise disjoint for all $g \in G$.

Example 18





[4] Images generated by Sonnet 5.

Even actions (cont.)

Theorem 19

If $p : Y \to X$ is a connected cover, then the action of $Aut(Y|X)$ on Y is even.

Proof.

Suppose that $y \in Y$, and let $p(y) = x$. Let U be a connected open neighbourhood of x such that $p^{-1}(U) \cong \coprod U_i$. One of these U_i must contain y, let it be V_0.
Now suppose that $\sigma \in \mathrm{Aut}(Y|X)$. Then $\sigma \cdot U_0 = U_i$ for some i by definition of a cover automorphism. By the Critical Lemma, distinct elements of the group gives different U_i. So the action is even. □

Even actions (cont.)

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Quotient covers

Theorem 20

If G is a group acting evenly on a connected space Y, the projection $p_G : Y \to G\backslash Y$ turns Y into a cover of $G\backslash Y$.
The map p_G is surjective and moreover each $x \in G\backslash Y$ has an open neighbourhood of the form $V = p_G(U)$ with a U as in Definition 2.1.6. This V is readily seen to satisfy the condition of Definition 2.1.1.

Quotient covers (cont.)

Theorem 21

If G is a group acting evenly on a connected space Y, then $Aut(Y|G\backslash Y) = G$.

Proof.

Notice first that we may naturally view G as a subgroup of $Aut(Y|(G\backslash Y))$. Now given an element ϕ in the latter group, look at its action on an arbitrary point $y \in Y$. Since the fibres of p_G are precisely the orbits of G we may find $g \in G$ with $\phi(y) = gy$. By the Critial Lemma we get $g = \phi$. □

Quotient covers (cont.)

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The Galois Correspondence

Galois covers

Definition 22

Assume that X is connected.
A cover $p : Y \to X$ is said to be *Galois* if the map $Y \to \mathrm{Aut}(Y|X)\backslash Y$ is an isomorphism.

Fiber transitivity

Theorem 23

Assume that X is connected.
A cover $p : Y \to X$ is Galois if and only if $Aut(Y|X)$ acts transitively on every fiber (in fact, only one fiber).

Proof.

Indeed, the underlying set of $Aut(Y|X)\backslash Y$ is by definition the set of orbits of Y under the action of Aut(Y —X) and so the map p is one-to-one precisely when each such orbit is equal to a whole fibre of p, i.e. when Aut(Y —X) acts transitively on each fibre. □